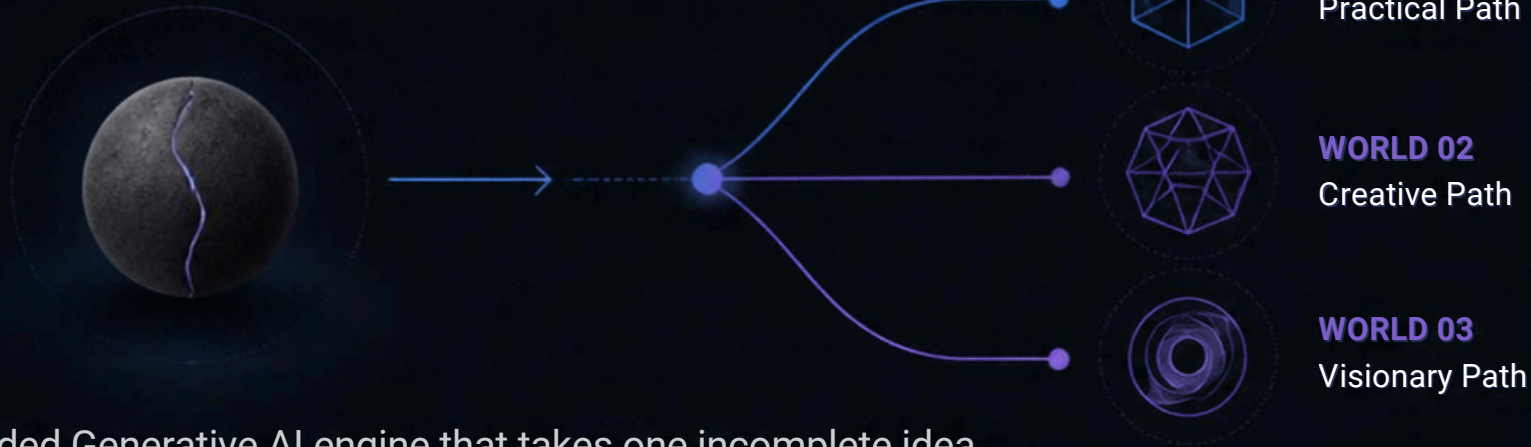


SEED UNFOLD

One small idea. Multiple possible worlds. One traceable solution.



A human-guided Generative AI engine that takes one incomplete idea, exposes its hidden intent and possibilities, generates **exactly three distinct solution or creative worlds**, and progressively unfolds the chosen direction into a **connected, persistent and traceable solution or universe**.



THEME

**Tattva 2 –
Forms hidden in formless**



DOCUMENT PURPOSE

**Single source of truth for product,
architecture, MVP scope,
implementation plan and judge demo.**

01 / Product thesis

Executive summary

Seed Unfold transforms **one incomplete seed** into understood intent, **three distinct worlds**, a **human-selected direction**, and a **connected persistent output**.

Who it serves

- Students
- hackathon builders and teams
- web/software developers
- product designers
- startup founders and early-stage innovators
- writers
- indie game developers
- animators and small creative studios
- educators
- creative/product teams

The problem

People have rough ideas but struggle to explore alternatives, structure them, preserve consistency across many AI outputs, and explain how a final result relates to the original idea. Typical workflows require repeated prompting across disconnected AI tools.

MVP boundary

-  **seed input**
-  **AI seed understanding**
-  **persistent Seed DNA**
-  **three worlds**
-  **human choice**
-  **5-7 visible unfolding stages**
-  **structured solution/universe**
-  **traceability**
-  **persistence**
-  **refine/regenerate**
-  **public deployment**
-  **fallback demo assets**

Core USPs

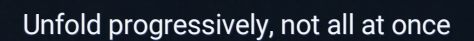
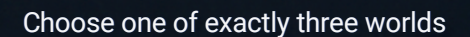
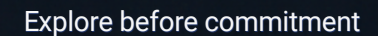
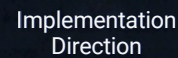
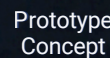
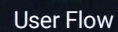
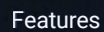
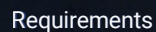
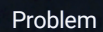
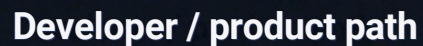
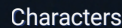
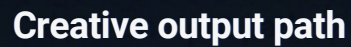
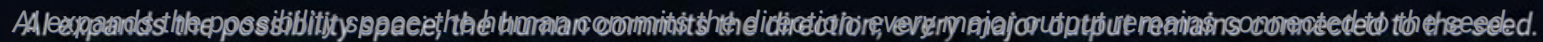
1  **Progressive Unfolding**
Seed → Intent → Context/Problem → Possibilities → Choice → Solution/Universe → Prototype/Media.

2  **One Seed, Multiple Worlds**
exactly three distinct directions before commitment.

3  **Seed DNA / Traceability**
major outputs retain lineage to the original seed.

4  **Human-guided AI**
AI explores, human chooses.

5  **Expandable Universe**
World → Chapters/Modules → Scenes/Features → Media/Prototype.

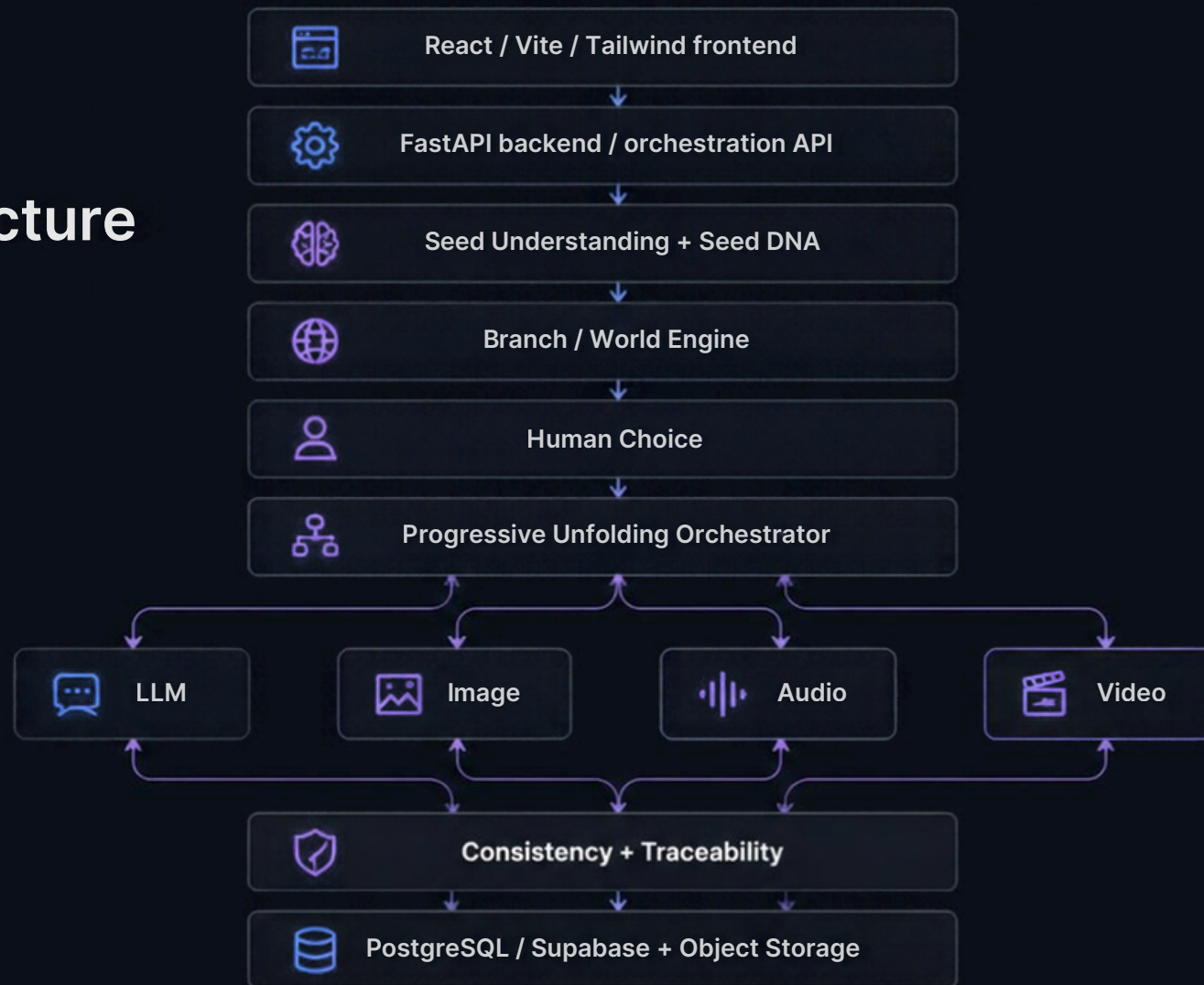




03 / System architecture



Provider abstraction keeps the core flow alive when media providers fail.



AI stack

Primary general-purpose LLM: Gemini for seed understanding, structured generation, branching and orchestration. TinyStories-style models are only an optional experiment for simple children's-story generation, never the main model. Optional components: vision model, image generation, TTS, music/audio generation, video generation, embeddings / pgvector.



Tech stack

Frontend: React + Vite + Tailwind CSS + Framer Motion | **Backend:** Python + FastAPI | **Data:** PostgreSQL / Supabase, optional pgvector / embeddings | **Storage:** Supabase Storage or equivalent | **Deployment:** Vercel / Netify plus Render / Railway / Cloud Run or equivalent

04 / Seed DNA, traceability & data

Seed DNA



Original Intent Users Context Constraints Entities Tone / Selected Lineage Seed

i Every downstream generation receives the relevant Seed DNA.

Traceability



i A 'Why?' interaction explains source metadata; it does not expose hidden chain-of-thought.

Data model

	Project	top-level workspace
	Seed	original idea or problem
	SeedDNA	structured interpretation and constraints
	Branch	one of the three possible worlds
	Selection	human choice and rationale
	Output	generated artifact or milestone
	TraceNode	lineage relationship
	GenerationJob	asynchronous generation state

API surface

<code></></code>	POST /api/projects
<code></></code>	POST /api/projects/{id}/seed
<code></></code>	POST /api/projects/{id}/understand
<code></></code>	POST /api/projects/{id}/branches
<code></></code>	POST /api/projects/{id}/select
<code></></code>	POST /api/projects/{id}/unfold
<code></></code>	POST /api/outputs/{id}/refine
<code></></code>	POST /api/projects/{id}/regenerate
<code></></code>	POST /api/projects/{id}/branch
<code></></code>	GET /api/projects/{id}
<code></></code>	GET /api/projects/{id}/trace
<code></></code>	

05 / Build plan, reliability & judge demo



Consistency + reliability

Every generation combines Original Seed + Seed DNA + Selected World + Existing World Facts / Characters + Current Output Goal.

	LLM inconsistency	→	schemas / validation
	Similar branches	→	diversity constraints
	Context drift	→	persistent world state
	Provider failures	→	abstraction / fallback
	Cost	→	optional media + caching
	Latency	→	parallel jobs / progressive status
	Invalid JSON	→	validation / repair
	Overbuilding	→	strict MVP



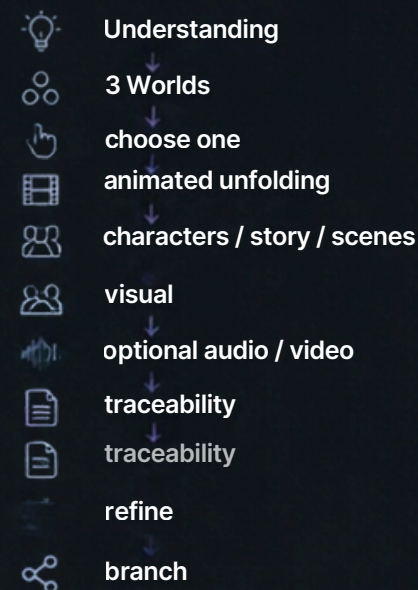
24-hour implementation plan

0-1	architecture
1-2	UI
2-4	seed engine
4-6	worlds
6-7	choice
7-9	unfolding
9-13	generation
13-14.5	consistency
14.5-16	traceability
16-17	persistence
17-18	refine / branch
18-19	hardening
19-20	deployment
20-21	demo mode
21-22	polish
22-23	judge simulation
23-24	freeze



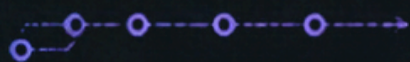
Judge demo

"A child discovers a forgotten city under the ocean."



Impact & roadmap

More accessible idea development, deliberate exploration; less disconnected prompting; stronger explainability; faster prototyping before production investment.



Voice / image seeds · long-form chapters / scenes · character memory · continuity checking · collaboration · exports · sharing · specialized agents · open-source / local model providers

One small idea. Multiple possible worlds. One traceable solution.

